

Manual End to End Test Cases

Use this document to record the results of manual end-to-end testing for the integrated Record Management System. Complete the test date, tester and build or commit fields before starting.

Test date: _26-09-20_ Tester: _Dongik Jung_ Build or commit: _f0494d83_

Running the Application

From the project root folder, run python main.py. Wait for the Record Management System window to open. Close it using Exit or the window close button so records are saved.

Test Data and Preconditions

- Start with a clean test data file or record the existing records. If project data must be retained, back up src/record/records.json before testing.
- Use Client ID 101, Airline ID 201 and Flight ID 301 unless an ID is already in use.
- Use the date format YY-MM-DD HH:MM, for example 26-09-20 14:30.
- Use Client data: Alex Smith, 10 River Road, Luton, Bedfordshire, LU1 1XX, UK, 07123456789.
- Use Airline data: Northwind Air. Use flight route: Luton to Dublin.

Manual Test Cases

ID	Scenario	Preconditions and test data	Steps	Expected result	Result
MTC-01	Load existing records on startup	records.json contains at least one known record.	Start the application and use the relevant display or search function.	Previously stored records are available in the GUI with their saved values.	Pass / Fail / Notes: <u>_Pass_</u>
MTC-02	Create a valid Client	No Client with ID 101 exists.	Open Client creation. Enter the valid Client data and save.	Client is created, appears in the GUI and a success message is shown.	Pass / Fail / Notes: <u>_Pass_</u>
MTC-03	Reject Client with missing required data	Use a new positive Client ID.	Leave Name or Address Line 1 blank and save.	The GUI shows a clear validation message. No record is created.	Pass / Fail / Notes: <u>_Pass_</u>
MTC-04	Reject duplicate Client ID	Client ID 101 exists.	Attempt to create another Client using ID 101.	The GUI rejects the duplicate ID. The original Client remains unchanged.	Pass / Fail / Notes: <u>_Pass_</u>
MTC-05	Search and display a Client	Client ID 101 exists.	Search for Client ID 101 and open the result.	The displayed record contains the correct Client fields and values.	Pass / Fail / Notes: <u>_Pass_</u>
MTC-06	Update a Client	Client ID 101 exists.	Change the City to Bedford and save the update.	The updated value is shown. Other Client values remain unchanged.	Pass / Fail / Notes: <u>_Pass_</u>
MTC-07	Delete a Client	A Client not referenced by a Flight exists.	Select the Client and use the delete operation.	The record is removed and no longer appears in a search result.	Pass / Fail / Notes: <u>_Pass_</u>
MTC-08	Create a valid Airline	No Airline with ID 201 exists.	Open Airline creation. Enter ID 201 and Northwind Air, then save.	Airline is created, displayed and confirmed by a success message.	Pass / Fail / Notes: <u>_Pass_</u>
MTC-09	Reject invalid Airline data	Use a new positive Airline ID.	Leave Company Name blank and save.	The GUI shows a validation message. No Airline record is created.	Pass / Fail / Notes: <u>_Pass_</u>
MTC-10	Reject duplicate Airline ID	Airline ID 201 exists.	Attempt to create another Airline using ID 201.	The duplicate is rejected and the existing Airline remains unchanged.	Pass / Fail / Notes: <u>_Pass_</u>

ID	Scenario	Preconditions and test data	Steps	Expected result	Result
MTC-11	Search and display an Airline	Airline ID 201 exists.	Search for Airline ID 201 and open the result.	The displayed record contains the correct Airline ID and Company Name.	Pass / Fail / Notes: _Pass_
MTC-12	Update an Airline	Airline ID 201 exists.	Change Company Name from Northwind Air to Northwind Airlines and save.	The updated Company Name is displayed. The Airline ID remains unchanged.	Pass / Fail / Notes: _Pass_
MTC-13	Delete an Airline	An Airline not referenced by a Flight exists.	Select the Airline and use the delete operation.	The record is removed and no longer appears in a search result.	Pass / Fail / Notes: _Pass_
MTC-14	Create a valid Flight	Client ID 101 and Airline ID 201 exist. Flight ID 301 is unused.	Enter Flight ID 301, Client ID 101, Airline ID 201, 26-09-20 14:30, Luton and Dublin. Save.	Flight is created and displayed with the entered values.	Pass / Fail / Notes: _Pass_
MTC-15	Reject duplicate Flight ID	Flight ID 301 exists.	Attempt to create another Flight using Flight ID 301.	The GUI rejects the duplicate. The existing Flight remains unchanged.	Pass / Fail / Notes: _Pass_

ID	Scenario	Preconditions and test data	Steps	Expected result	Result
MTC-16	Reject invalid flight date format	Client ID 101 and Airline ID 201 exist. Use a new Flight ID.	Enter 2026-09-20 14:30 or another invalid format and save.	The GUI shows a clear validation message. No Flight is created.	Pass / Fail / Notes: _Pass_
MTC-17	Reject missing referenced record	Use a new Flight ID and one non-existent Client ID or Airline ID.	Enter otherwise valid Flight data and save.	The GUI rejects the Flight and identifies that the referenced record is missing.	Pass / Fail / Notes: _Pass_
MTC-18	Search and display a Flight	Flight ID 301 exists.	Search for Flight ID 301 and open the result.	Correct Flight data is displayed. The stored data is not changed by searching.	Pass / Fail / Notes: _Pass_
MTC-19	Update a Flight	Flight ID 301 exists.	Change End City from Dublin to Cork and save.	The updated route is displayed and the record remains valid.	Pass / Fail / Notes: _Pass_
MTC-20	Delete a Flight	Flight ID 301 exists.	Select the Flight and use the delete operation.	The Flight is removed and no longer appears in a search result.	Pass / Fail / Notes: _Pass_
MTC-21	Confirm persistence after creation	Create a valid Client, Airline and Flight.	Close the application normally. Restart it and search for all three records.	All records remain available with the saved values.	Pass / Fail / Notes: _Pass_
MTC-22	Confirm persistence after update and deletion	Records exist and have been changed or deleted through the GUI.	Close the application normally. Restart it and repeat the relevant searches.	The update is retained and the deleted record does not return.	Pass / Fail / Notes: _Pass_
MTC-23	Confirm failed operations do not change data	Record the current details of a valid record.	Attempt an invalid create or update, then search for the original record.	The original record remains exactly as it was before the failed operation.	Pass / Fail / Notes: _Pass_